ICDETECTOR EXPORTS · PLAIN-LANGUAGE REPORT FOR NON-SPECIALISTS

Plain-language report — Patrones sintéticos

What your phone's data say about possible fake base stations, explained without jargon. The details are in the full technical report.

Case	Patrones sintéticos — Reproduce la forma de los patrones de un expediente real, con datos inventados
Phone	Teléfono sintético (sin dispositivo real) (declared by whoever created the case)
Data	2026-09-24 – 2026-10-02 · 5 ICdetector files
Data origin	Synthetic test data
Generated	2026-10-02 08:05:00 UTC
Report language	English (en-GB); the language only changes the presentation, not the analysis

> **Notice**
> Synthetic test data: they do not describe any real phone, network or place.

1. What can the data tell us about possible fake base stations?

> **Status: Observations that need more data**
> No observation allows an attack to be attributed. One observation cannot be clarified with these data and it is worth gathering more information.

A fake base station (sometimes called an «IMSI catcher») is a device that pretends to be a legitimate mobile base station; it can be used to identify or track phones and, depending on the system and the network, to carry out other attacks. This report cannot confirm or rule out its presence: it assesses whether the data give reasons to investigate it.

2. Which observations deserve attention, and why?

In figures: 13 ICdetector records grouped into 9 episodes; 8 episodes checkable against the saved measurements. A record is a note made by ICdetector; an episode groups the notes about the same cell and type that are close in time; it is checkable if CellCase can compare it with the measurements.

- A cell switched back and forth between two radio identifiers (on 29 September): more data needed. It affects how the cell identifies itself, something a fake base station could imitate; it can also have normal explanations.
- Frequency changes within the 4G network (on 25 September and 29 September, 9 times): insufficient evidence; this is common network behaviour, and its recurrence does not increase suspicion.
- Signal much stronger than usual (on 27 September, once): insufficient evidence; it can have normal explanations.
- New cell at a usual place (on 27 September, once): insufficient evidence; it can have normal explanations.
- In addition, Radio recorded degraded service states with no other warning nearby (8 out of service); they can have normal explanations.

The technical report marks E2, E3, E8, E9 for technical review because they recur or are visible in the measurements. This is technical interest, not a reason to suspect spoofing.

3. What can you do next?

- A cell switched back and forth between two radio identifiers: if you go back to that place, leave ICdetector running; if it opens a forensic case, export it and add it to this case.
- For the other observations, these data alone do not justify urgent action: keep ICdetector running as usual.
- If a specific situation worries you (for example, at a particular place or date), keep this report and the technical one and consult a person specialised in telecommunications security.

4. The observations, one by one

1. A cell switched back and forth between two radio identifiers — More data needed

What happened

- On 29 September, ICdetector noted that a cell (C6) was changing its radio identifier. The saved records show that it moved to another identifier and back to the previous one.
- It did not reach ICdetector's alarm threshold: it was noted, but ICdetector did not warn.

Which explanations fit

- The phone may read data from the previous cell during a cell change and mix them for a moment.
- The operator may have reconfigured that cell, or several nearby cells share part of their identification.
- ICdetector watches for this pattern because a fake base station could imitate the identity of a real cell; a brief, isolated case can have normal explanations, and these data can neither rule it out nor confirm it.

What is missing to clarify it

- A detailed ICdetector record (forensic case) of that moment, with the neighbouring cells, and knowing whether it recurs at that place.

What to do

- If you go back to that place, leave ICdetector running; if it opens a forensic case, export it and add it to this case.

Technical details: E6 in the full technical report.

2. Frequency changes within the 4G network — Insufficient evidence

What happened	On 25 September and 29 September, the phone moved to another frequency of the same 4G network 9 times, with 4 cells (C3, C1, C2, C7). It almost always went from a higher frequency to a lower one. It did not reach ICdetector's alarm threshold: it was noted, but ICdetector did not warn. CellCase checked it in the saved measurements: the change is visible in the data.
Explanations	Operators change frequencies often to spread the traffic. Low frequencies reach further and get into buildings better, so the phone may drop to them indoors or far from the station. Recurrence in the same cells and places may reflect how the operator has configured that area.
What is missing	For these isolated frequency changes, no priority additional check is identified. It is worth reviewing whether they coincide with other anomalies.
What to do	These data alone do not justify urgent action. Keep ICdetector running as usual.

Technical details: E1, E2, E3, E7, E8, E9 in the full technical report.

3. Signal much stronger than usual — Insufficient evidence

What happened On 27 September, the signal of a cell (C4) was quite different from what that phone usually records with that cell, once. It did not reach ICdetector's alarm threshold: it was noted, but ICdetector did not warn. CellCase checked it in the saved measurements: the change is visible in the data.

Explanations The signal changes a lot depending on where the phone is: inside or outside a building, on a high floor or in a basement. The «usual» reference mixes different places and times, so a one-off difference is to be expected.

What is missing Repeated measurements at the exact same spot to have a comparable reference.

What to do These data alone do not justify urgent action; if it recurs often at the same place, generate the report again.

Technical details: E4 in the full technical report.

4. New cell at a usual place — Insufficient evidence

What happened On 27 September, at a place where the phone is usually found, ICdetector saw a cell it had not recorded for that place. It did not reach ICdetector's alarm threshold: it was noted, but ICdetector did not warn.

Explanations The operator may have activated or changed a cell. The phone may have connected to a slightly more distant cell because of coverage changes.

What is missing Seeing whether that cell keeps appearing at that place over the next few days.

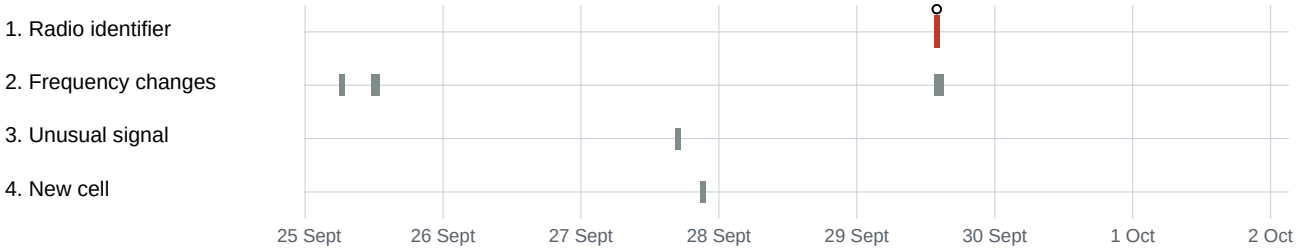
What to do These data alone do not justify urgent action; if it recurs often at the same place, generate the report again.

Other sources Topology noted a change between those same cells during those minutes (with a margin of 1 min). It may be the same change noted twice by ICdetector; either way it is not an independent confirmation. The history has no notes during those minutes, so it cannot be checked against it.

Technical details: E5 and the section «Agreements and differences between sources» in the full technical report.

Figure 1. When each observation happened

local time in Europe/Madrid (declared assumption)



Each mark is a moment noted by ICdetector, within the period covered by the data. In red, observations that need more data; in grey, those that give no indications. It shows whether they cluster on particular days; the number of marks does not measure severity.

5. What this report cannot say

- It cannot confirm or rule out a fake base station: CellCase does not yet have tested criteria to flag indications that require specialist investigation, which is why this report does not use that category.
- ICdetector notes many warnings that do not reach its alarm threshold; its checks are not calibrated and can be triggered by normal explanations.
- An isolated record does not say how long what was observed lasted, and the absence of warnings does not prove that there has been no problem at all.
- The phone, the time zone and the origin of the data are what the person who created the case declared.
- Data are missing for some analyses (1); the technical report gives the details.

Data analysed: 5 ICdetector files (observation history, radio: service state events, topology, stable-site, geometry). Cells are named C1, C2… as in the full technical report, which gives their identification, the measurements, the rules (episodes/6) and the fingerprint of each file.